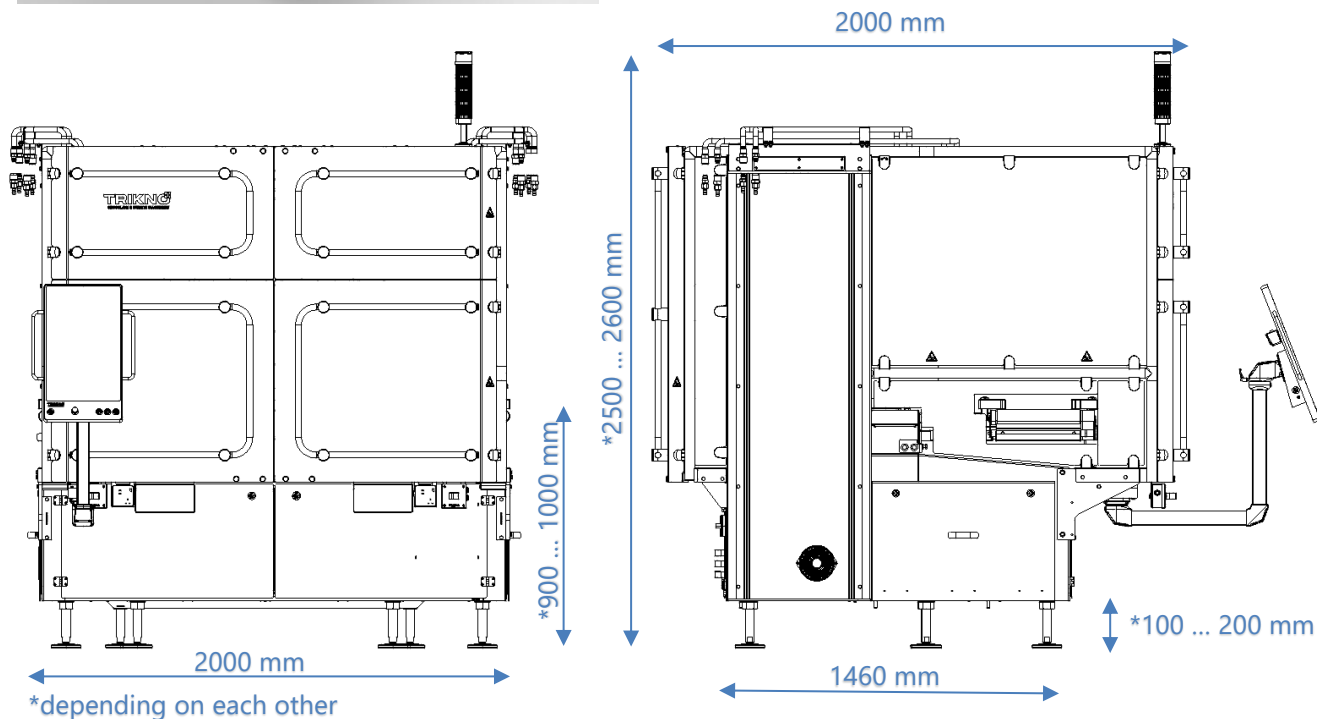
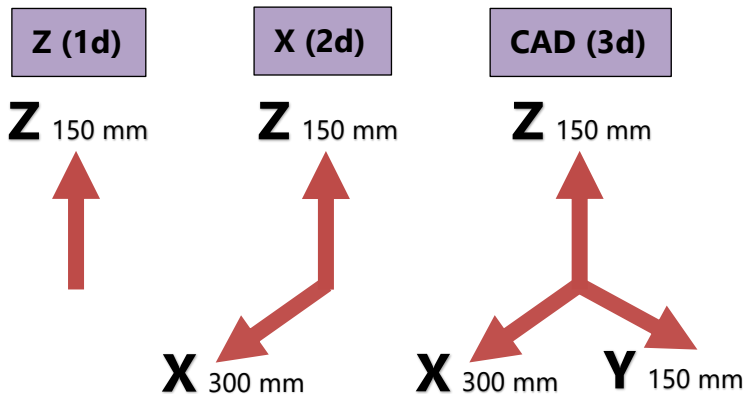
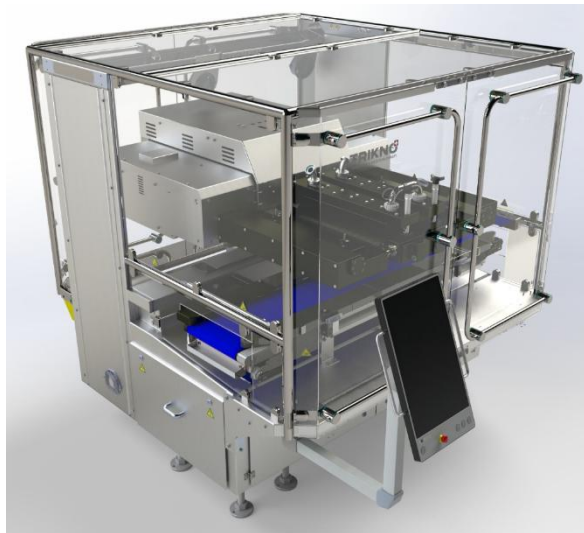




Technical Data Sheet TCM Alpha-Nut

Machine with proven single-rotary piston system, powered entirely by electric servo motors.
Wide range of applications.



Single	9, 12, 16, 18, 24	9, 12	Optional
Boxer	9/18, 12/24, 16/32, 18/36, 24/48	9/18, 12/24	For detailed information, see variants further down in the document
Liner width	45 mm	90 mm	
Piston distance	22.5/45 mm	90 mm	
Piston stroke	70 mm		
Conveyor length	2'000 mm		... 3'750 mm
Mould widths	261 ... 412 mm		... 517 mm
Weight	2'000 kg	2'500 kg	



Features

- Head movement by electric servo motors possible as single-axis (vertical: Z) or two-axis (vertical and belt direction: X) or as triple axis (CAD)
- High-precision single rotary piston system with identical dosing volumes (proven for more than 30 years – Single Piston System)
- No mixing between the pistons possible (1 piston = 1 nozzle)
- Machine designed for industrial application
- Quick changeover of depositing tools (pistons/liner/nozzle plates) without additional tools
- Number of pistons individually selectable (easy activation / deactivation by using individual couplings)
- Easy cleaning of the dosing machine without additional tools (Stainless steel or aluminium in ALTEF® coating – for pH values 4 ... 8)
- Easy cleaning of the pistons and cylinder liner (no tools necessary)
- Movable touchscreen HMI for machine operation
- Depositing head in free-arm design for easy operation / cleaning / adjustment
- Fully digital machine control
- Remote access on PLC, HMI, servo drives possible
- Software designed for maximum accuracy and flexibility
- One Shot software already installed on every machine
- Double-walled hopper on both sides

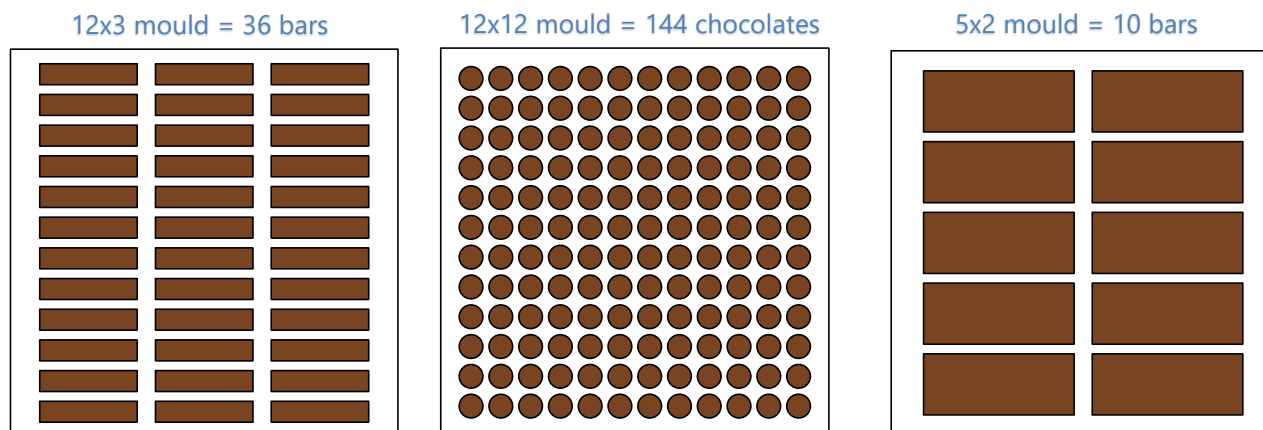
Options

- Piston lubrication system for one or both piston sides (drip lubrication)
- Slider plate for big inclusions on big nozzles
- Different agitator systems for one or both hoppers
- Triple Shot (1 filling / 2 shells OR 2 fillings / 1 shell)
- Quatro Shot (2 fillings / 2 shells)
- Control of following machine (e.g. vibration table) possible via control panel
- Y 300 mm (Default 150)
- Belt running direction from left to right or right to left
- RFID for mould recognition
- Hybrid machine (1 side chocolate, 1 side nut) or as nut machine

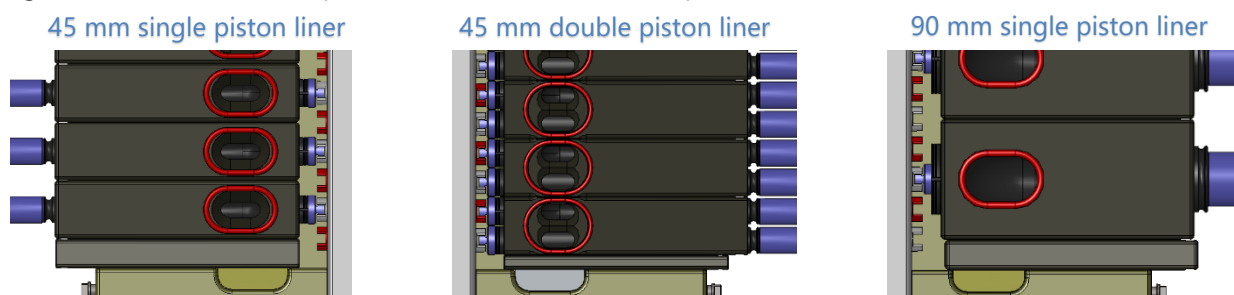


Combination options with 45 mm cylinder liners

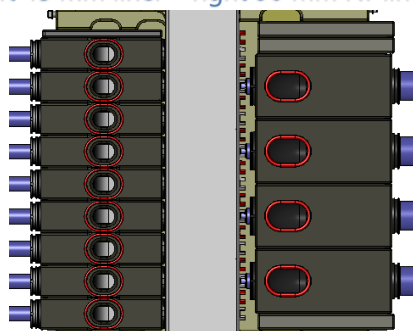
The Alpha Nut is designed in such a way that single 45 mm piston liner, 45 mm double piston liner and 90 mm single piston liner can be combined as desired. Here are shown examples with moulds in dimensions 500 mm x 500 mm.



The alignment of the off-centre pistons, dummies can serve as placeholders.



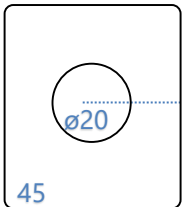
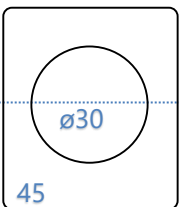
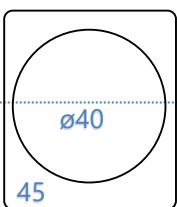
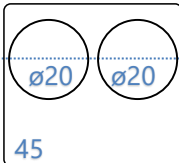
Example of a Hybrid-solution
left 45 mm liner – right 90 mm NI-liner



Piston- ø mm	Only Inclusions		OneShot or small inclusions			Piston volume	
	Nut 45	90	Nut-i 45/2	45	90	ml	g
20			X	X		22.0	26.4
30	X			X		49.5	59.4
40	X			X		88.0	105.6
50		X			X	137.4	164.9
60		X				197.9	237.5
70		X				269.4	323.3



NI (small inclusions) Piston dimensions (also possible as Hybrid)

Liner width 45 mm Stroke 70 mm				
Piston - Ø	20 mm	30 mm	40 mm	2x20 mm
Volume	22 ml 26.4 g	49.5 ml 59.4 g	88 ml 105.6 g	2x22 ml 2x26.4 g
Recommendations	One Shot with 2 ... 3 mm Inclusions = up to 15% share Cluster or solid with 4 ... 5 mm Inclusions = up to 10% share	Cluster and bars with 4 ... 5 mm Inclusions = up to 12% share	Bars with 7 ... 8 mm Inclusions = up to 22% share Bars with 9 ... 11 mm Inclusions = up to 25% share	One Shot with 2 ... 3 mm Inclusions = up to 15% share Cluster or solid with 4 ... 5 mm Inclusions = up to 10% share

Example 1: 100g bar with big inclusions

TCM Alpha Nut 24/48

Piston NI 2x5-45/1-Ø40 (2 pistons for one bar)
10 bars, 15 strokes/min and 60 min/h results in:

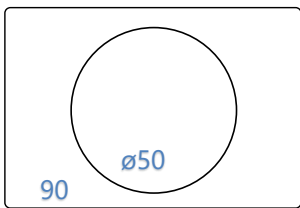
➤ **9'000 bars/h or 900 kg/h**

Example 2: 10g One Shot Chocolates with inclusions

TCM Alpha Nut 24/48

Piston NI 2x12-45/2-Ø20 (1 piston for one chocolates)
24 chocolates, 20 strokes/min and 60 min/h results in:

➤ **28'800 chocolates/h or 288 kg/h**

Liner width 90 mm Stroke 70 mm	
Piston - Ø	50 mm
Volume	137.4 ml 164.9 g
Recommendations	Bars with 7 ... 8 mm Inclusions = up to 28% share Blocks with 9 ... 11 mm Inclusions = up to 30% share

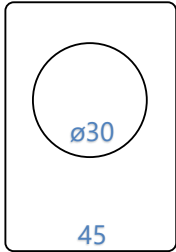
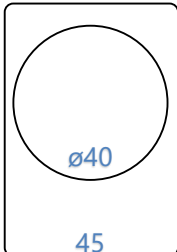
Example 3: 100 g bars with big inclusions

TCM Alpha Nut 24/48

Pistons NI 2x5-90-1-Ø50 (1 piston for one bar)
10 bars, 15 strokes/min and 60 min/h results in:

➤ **9'000 bars/h or 900 kg/h**

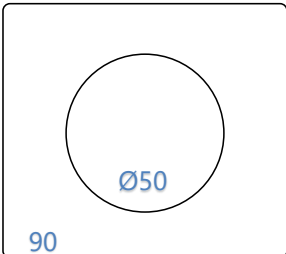
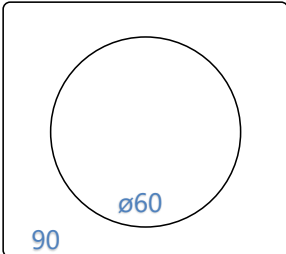
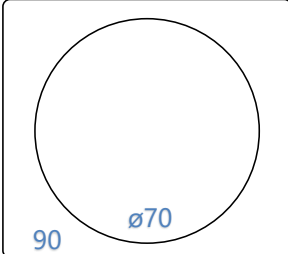
N (big inclusions) piston dimensions (not possible as Hybrid)

Liner width 45 mm Stroke 70 mm		
Piston - Ø	30 mm	40 mm
Volume	49.5 ml 59.4 g	88 ml 105.6 g
Recommendations	Cluster and bars with 7 ... 8 mm Inclusions = up to 12% share	Bars with 7 ... 8 mm Inclusions = up to 22% share Bars with 9 ... 11 mm Inclusions = up to 25% share

Example 4: 150g bars with inclusions (mould 500x500)

TCM Alpha Nut 12/24 with pistons NI 2x5-45-1-Ø40 (2 pistons for one bar)
10 bars, 15 strokes/min and 60 min/h results in:

➤ **9'000 bars/h or 1'350 kg/h**

Liner width 90 mm Stroke 70 mm			
Piston- Ø	50 mm	60 mm	70 mm
Volume	137.4 ml 164.9 g	197.9 ml 237.5 g	269.4 ml 323.3 g
Recommendations	Bars with 7 ... 8 mm Inclusions = up to 28% share Bars with 9 ... 11 mm Inclusions = up to 30% share	Bars with 7 ... 8 mm Inclusions = up to 30% share Bars with 9 ... 11 mm Inclusions = up to 33% share	Bars with 7 ... 8 mm Inclusions = up to 33% share Bars with 9 ... 11 mm Inclusions = up to 38% share

Example 5: 200 g blocks with whole nuts (mould 500x500)

TCM Alpha Nut 12/24 with pistons NI 2x5-90-1-Ø70 (1 piston for one block)
10 bars, 15 strokes/min and 60 min/h results in:

➤ **9'000 blocks/h or 1'800 kg/h**

Applications



- Bars without filling
- Bars with small inclusions up to 3 mm
- Bars with whole nuts



- Filled chocolates in 2 colours (One Shot)
- Filled candy bars in 2 colours (One Shot)
- Filled chocolates in 3 colours (Triple Shot)
- Filled chocolates in 4 colours (Quatro Shot)
- Filled chocolate bars (One Shot Curtain)



- Free forms without filling
- Free forms with fillings
- Free forms with two fillings



- Fillings for Macarons
- Fillings for Alfajores

